

The Design and Implementation of New Vehicle Black Box Using The OBD Information

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Abstract—In this paper, the new Vehicle Black Box, which is named E-Black Box (Enhanced Black Box), is designed and developed. The E-Black Box is the combination of all the advantages of previous Black Boxes and Event Data Recorders (EDRs). The E-Black Box communicates with Electronic Control Units (ECUs) to read the status of vehicle via OBD port. It is equipped with camera that records the video in front of the vehicle. As the common vehicle Black Box, the E-Black Box also has Global Positioning System (GPS) for reading the current latitude and longitude of the vehicle point. The 9 Degrees of Freedom (DOF) of inertial sensor, which is incorporated triple-axis gyro sensor, triple-axis accelerometer sensor and triple-axis magnetometer sensor, is integrated with E-Black Box to read the velocity, acceleration and orientation of vehicle after which we analyze the stability of vehicle during the travel. After collecting and synchronizing all data, the E-Black Box saves them in Secure Digital (SD) Card.

Keywords: EDR, OBD, Vehicle Black Boxes.

I. INTRODUCTION

The black box is introduced firstly in 1953 by David Warren [1] for the purpose of facilitating the investigation of an aircraft accident or incident. It was not only able to record all important data of the flight instruments but also to record the cockpit voices during the flight. It became a mandatory feature in 1960, when the Federal Aviation Administration required all commercial planes to carry them. Until in 2004, in efforts to establish the uniform scientific crash data needed to make vehicle and highway transportation safer and reduce fatalities, the first universal standard for Motor Vehicle Event Data Recorders (MVEDRs) [2] was launched by IEEE Standards Association. It is a device installed in automobiles to record information related to vehicle crashes or accidents. MVEDRs are also called Event Data Recorders (EDRs) or Vehicle Black Boxes, but there are a few differences between them that we will show below. Information from these devices can be collected after a crash and analyzed to help determine what the vehicle were doing before, during and after the crash or event.

In the market today, the number of vehicles equipped with black boxes is increasing. Most of common vehicle black boxes which are equipped with a high resolution camera and wide lens that can capture clear and color images in the front of the vehicle and save them on memory. They also have the Global Positioning System (GPS) for navigation purposes that

provides location and time information anywhere on the Earth. Depending on the size of the memory card in vehicle black boxes; they can save amount of information of the journey locations and video record captured by GPS and camera, respectively. These can be used in evidence should a court case occur. Those are the popular techniques on the common vehicle black boxes in the market.

The EDRs are developed for collecting, recording, storing and exporting all data related to vehicle crashes or accident. These days most vehicles are equipped with EDRs, unfortunate EDRs functions are different from different manufactures. But the basic EDRs must recognize engine events such as engine temperature, manifold pressure data, engine speed, airbag status, Vehicle Identification Number (VIN), pedal, ABS sensors, etc. When fault-related events occur or events triggered, the data are collected and written to memory. The data later can be downloaded using a computer to analyze and verify the status of vehicle during the events.

When motor vehicle accident happens, it takes the investigator to piece the puzzle and establish the sequence of occurrences. It is difficult to establish the exact cause of an accident if either information of the vehicle or video record and location are absent. As in [3], [4], [5], [6], we can see many advances on implementations of black boxes and EDRs. In [3] they integrated GPS with Standalone MVEDRs, with [5], [6] they implemented the Vehicle Black Box on Embedded System and Embedded Controller. In [4], the authors combine the black boxes and EDRs system to provide more accurate information regarding a vehicle accident. The information are provided base on On-Board Diagnostic II protocol (OBD-II) to record vehicle status data and MOST protocol for recording a vehicle's multimedia and navigation information. In this paper, we design and implement the vehicle black box with more advanced properties than the previous. We also name the new vehicle black box is E-Black Box which means Enhanced Black Box.

II. E-BLACK BOX SYSTEM

The E-Black Box works independently with all vehicle hardware equipment and customers install easily in any vendor car. The E-Black Box supports the OBD-II protocol for communication with ECUs on the vehicle. Therefore it can hold information of vehicle such as VIN, airbag status, engine speed, etc. The E-Black Box also has a camera that records

sights outside the cabin of the vehicle. Besides that, the E-Black Box is equipped with GPS module and 9 Degrees of Freedom (DOF) of inertial sensor which includes triple-axis accelerometer, triple-axis gyro and triple-axis magnetometer sensor for holding the latitude, longitude of the current vehicle point, accelerations and attitude (yaw/pitch/roll) of the vehicle during the travel. Following the 1616 IEEE Standard for MVEDRs [2], this information which are collected by E-Black Box has to record in 10 seconds before and 10 seconds after accident happens. However, the E-Black Box records continuously when vehicle starts working, it synchronizes all information by real-time clock in every 100 milliseconds and then save them in memory card. By this way, the inspector can understand exactly what happen in every 100 milliseconds. The E-Black Box stops recording after 10 seconds of accident or the vehicle stops working. Depending on the size of memory card is inserted inside E-Black Box; huge data for several days can be stored. If the memory is overload, the oldest data will be renamed and overlap for the new ones. We also develop the software that the customer runs the simulation online and offline on E-Black Box. If active the online mode, the E-Black Box has to communicate with the computer via USB Port for showing the status vehicle such as temperature engine, speed engine, the journey vehicle, acceleration, orientation of vehicle and so more. The offline mode is the normal mode working of the E-Black Box. For running the offline simulation, the customer only inserts the SD Card which is inside the E-Black Box to the computer. After that, our software automatically reads and rebuilds all necessary data for offline simulation. It shows the report and simulations of the vehicle operation with the desired time. Another advanced function in E-Black Box is communication with the customer's smart phone via Bluetooth. In case of accident, the E-Black Box sends information packets about the vehicle status to customer's smart phone. The customer's smart phone searches the nearest service center, hospital or the garage around accident place and sends the messages of location and the status of vehicle to those places for asking help. Figure 1 is the block diagram of the E-Black Box.

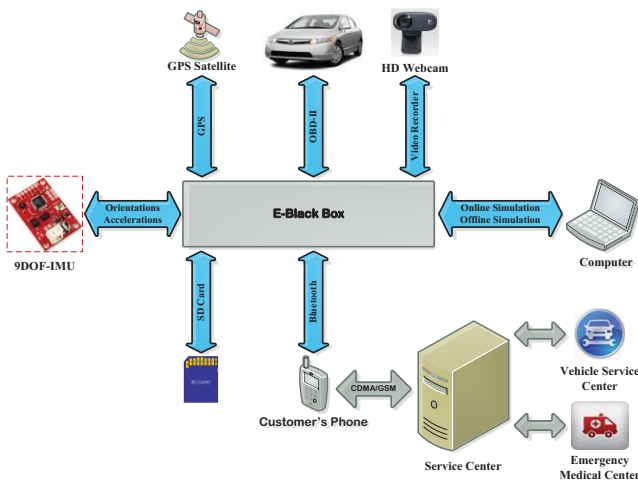


Figure 1: The Block Diagram of the E-Black Box.

III. THE DESIGN AND IMPLEMENTATION OF THE E-BLACK BOX

A. OBD-II

In this part, the On-Board Diagnostic II (OBD-II) system is reviewed. OBD-II is a set of standards for implementing a computer based system to monitor the performance of some of an engine's major components including those responsible for controlling emissions. In the early 1990s, Society of Automotive Engineers (SAE) and International Standardization Organization (ISO) issued a set of standards which described the interchange of digital information between ECUs and a diagnostic scan tool. All OBD-II compliant vehicles were required to use a **standard diagnostic connector** (SAE J1962), and communicate via one of the standard OBD-II **communication protocols** which are ISO 15765-4/SAE J2480 (CAN), SAE J1850VPW (GM Vehicles), SAE J1850PWM (Ford Vehicles), ISO 9141-2 (Asian, European, Chrysler Vehicles), ISO 14230-4 (Keyword Protocol 200), and SAE J1939. It was first introduced in the United States in 1994, and became a requirement on all 1996 and newer US vehicles. A large portion of the modern vehicle fleet supports OBD-II or one of its regional flavors (EOBD, JOBD, et al.)

OBD-II PIDs (On-Board Diagnostic Parameter IDs) are codes used to request data from vehicle. SAE standard J1979 [7] defines many standard PIDs, but manufactures also define more nonstandard PIDs specific to their vehicle. There is limited information available in the public domain for nonstandard PIDs. Therefore, we use the standard PIDs in this implementation because it is supported to almost vehicles. In this paper, the legislated OBD-II standard is ISO 15765-4/SAE J2480 which is communicated with ECUs via Control Area Network (CAN) protocol is implemented in order to hold the vehicle information such as: **engine load, coolant temperature, fuel pressure, engine rpm, vehicle speed, throttle position, fuel level input, barometric pressure, accelerator pedal position, engine oil temperature, mass airflow**, etc.

B. 9 Degrees Of Freedom Inertial Measure Unit

The 9 Degrees Of Freedom Inertial Measure Unit (9 DOF IMU) is an electronic device incorporates three sensors which are triple-axis gyroscope (IGT-3200 is triple-axis MEMS gyro IC with a full-scale range of $\pm 2000^\circ/\text{sec}$), triple-axis accelerometer (ADXL345 is a small, thin, low power, complete triple-axis accelerometer with a minimum full-scale range of $\pm 3g$) and triple-axis magnetometer (HMC5883L is a triple-axis digital magnetometer designed for low-field magnetic sensing with a digital interface, has a full-scale range of ± 8 Gauss and resolution of up to 5 milli-Gauss). IMUs are typically used to measure and report velocity, orientation, and gravitational forces on maneuver aircraft, unmanned aerial vehicles, spacecraft, satellites, and Landers. In this paper, the 9 DOF IMU adds an on-board processing system that provides solved heading, attitude and yaw information for vehicle. An estimation of orientation of IMU orientation in this paper is using a gradient descent algorithm in [8]. The implications of the low computational load and ability to operate at small sampling rates significantly reduces the hardware and power necessary for wearable inertial movement tracking, enabling

the creation of lightweight, inexpensive systems capable of functioning for extended periods of time. With the online simulation, the orientation of vehicle is plotted in a graph in every 100 milliseconds. It is very important and useful for the technician that if they want to analysis the stable of the vehicle in real-time. We read also the raw data of this sensor for understanding the acceleration, deceleration of the vehicle in any direction. In offline simulation, all information of 9 DOF IMU save as history of orientation, acceleration, deceleration of vehicle and synchronous with the real-time clock. After that we can rebuild the simulation in specific time bases on information collected in SD Card.

C. Camera Recording

We can find many cameras that are integrated in traditional Black Boxes. These cameras record continuously video in front, interior view of the vehicle. Therefore the camera in this case uses as the evidence if an accident occurs. It also acts as the security of the vehicle. The embedded system in E-Black Box is implemented with the Linux Operating System. Obviously, this operating system also supports the USB protocol that recognizes automatically any off-the-shelf webcam devices in the market. This is an advantage of the E-Black Box. Because with the USB ports we can plug in many webcams replace for the traditional camera of Black Boxes. Then we can record the video or capture the pictures in many areas such as front-side, inside, back-side, left-side, and right-side of the vehicle. In this paper, we use a video camera with 720p HD videos and 5-megapixel.

In online simulation mode, we use webcams as video recording. The application on this mode shows everything happen inside and outside the vehicle through webcams in real-time. In offline simulation mode, we program in multithread to manage each webcams. Because the size of the video file is real big; and the size of the SD Card is limited. Instead of recording the video, in offline simulation mode, one thread is triggered in every 10 millisecond to allow the webcam captures the picture. This method reduces the size the data save on SD Card. The application for offline simulation mode will makes the video file with 10 pictures per second revolution base on recovering the stored pictures in SD Card.

D. Embedded Systems

The embedded system of E-Black Box is implemented with Linux. Linux and most GNU software are licensed under the GNU General Public License (GPL) that allows everyone can freely download, develop and contribute. Our purpose is to make the E-Black Box free and open source software. We would like to share the implementation that everyone can develop and make E-Black Box to be more stable and high performance. The embedded system board is used in this paper is **Beagle Board-xM**. The Beagle Board-xM is a low cost, fan less single board computer based on TI's OMAP3 device family, with all of the expandability of today's desktop machines, but without the bulk, expense, or noise. It uses a TI OMAP3530 processor (ARM Cortex-A8 superscalar core ~ 600MHz paired with a TMS320C64x+ DSP ~ 430MHz and an

Imagination SGX 2D/3D graphics processor). Figure 2 shows the real E-Black Box that we design and implemented.

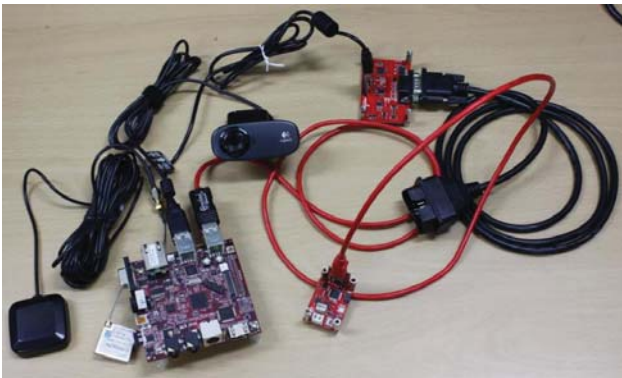


Figure 2: The E-Black Box

The function of embedded system of E-Black Box are collecting and monitoring all sensors and external device (GPS, 9 DOF IMU, OBD-II, Webcams, Bluetooth, etc.) in real-time, running offline or online simulation. Based on these functions, the embedded system structure of E-Black Box is divided into 3 layers and is shown in Fig. 3.

- The first layer is the device layer. This layer includes all sensors, external devices and signals from vehicle such as: Bluetooth from smart phone, GPS device, OBD-II device, SD Card, Webcams, 9 DOF IMU. The mission of this layer is collecting the information of vehicle for E-Black Box.
- The second layer is hardware layer. Hardware layer is the layer that implement with all protocol stacks are Bluetooth, CAN, USB, UART, SPI, GPIO, Ethernet that allow the sensors, external devices communicate with the third part.
- The third layer is the Operating System layer. The Operating System monitors all devices, sensors. It saves, analyzes all information and runs the simulation. We use the Linux operating system for E-Black Box. The Linux distribution for ARM processor is implemented in this paper is Ubuntu.

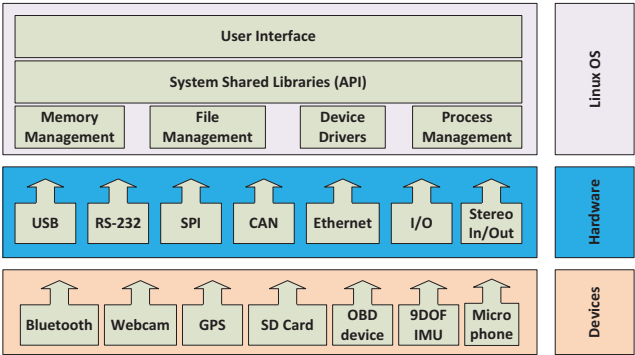


Figure 3: The Structure of E-Black Box System

IV. EXPERIMENTAL RESULTS

The collected data are used to reconstruct what happened inside the car and outside environment. The **insurance agent**, **police department**, **vehicle design**, **roadway design**, **automaker service center**, **emergency medical service**, **vehicle owner** or **government** needs all vehicle information related to car accident. This information is also useful to improve the quality of the vehicle, roadway, etc. The simulation in this paper has two modes: offline and online simulation.

Online Simulation: All information of the vehicle is showed in real-time. This mode is useful to analysis the status of vehicle during the operation. We can remote the E-Black Box and activate the online simulation mode. The online simulation supports the user interface to visualize all information via HDMI connector or we remote E-Black Box from another computer to show information. Figure 4 is the online simulation that we done with the ECUsim-2000 which is OBD-II ECU Simulator. The figure shows the orientation of the vehicle (yaw/pitch/roll) collect by 9 DOF IMU and OBD-II information (engine speed, vehicle speed, engine coolant temperature, voltage of oxy sensor, mass air flow sensor, and engine load).

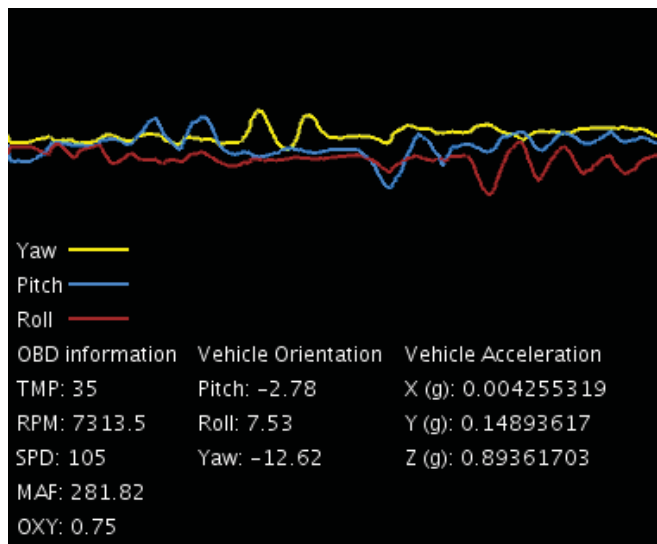


Figure 4: The online simulation with ECUsim-2000

Offline Simulation: The simulation in this mode run by reconstructed all stored data in SD Card. The stored information in SD Card is saved in specify folder with each sensor and devices. We develop the software that can read and rebuild stored information. This way allows us to see what happened with the vehicle in any specify time.

V. CONCLUSIONS

In this paper, the design and implementation of the new Vehicle Black Box is proposed. The new Vehicle Black Box is integrated GPS, Webcams, OBD-II, and 9 DOF IMU in embedded system for high performance and stability. With advantages of embedded system, E-Black Box can be implemented with more devices and become more intelligent. The simulation goes together with E-Black Box that help the vehicle customer, insurance agent, police, vehicle design, roadway design and emergency medical service center, government to analysis the vehicle as well as the driver behavior and roadway. With the developments of the embedded processor, the future work of this paper is implemented with other algorithms such as: tracking the road, notice the driver if have any obstacles on the road etc. We also propose the implementation E-Black Box base on Embedded System that can support the high performance for E-Black Box. The Beagle-XM board, which is selected in this implementation, uses the TI AM3358/9 SoC based on an ARM Cortex-A8 processor core using the ARMv7-A architecture. It has many features that satisfy all our requirements and is developed together Linux Operating System for collecting and analyzing data as well as communicating with other devices.

VI. REFERENCES

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